

SALAJ CHOUDHARY

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SRM Institute of Science and Technology, Trichy, India

Education

SRM Institute of Science and Technology

2023 – 2027

B.Tech in Information Technology

CGPA: 9.33/10

Technical Skills

Languages: Java, Python, JavaScript, C, C++

Frontend: React.js, React Native, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend: Node.js, Spring Boot, REST APIs, Authentication & Authorization

Databases: MySQL, MongoDB, Firebase Realtime Database

Tools: Git, GitHub, Postman, Vite, Expo

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Multithreading

Experience

Naya Sawera NGO – Software Developer(Client Engagement)

Jun 2025 – Present

- Developing and maintaining a production-ready web platform supporting donor management, volunteer onboarding, and community outreach initiatives.
- Collaborating directly with NGO stakeholders to gather requirements and translate business needs into scalable software solutions.
- Implementing secure donation workflows, volunteer registration systems, authentication mechanisms, and robust form validation processes.
- Building reusable React components and responsive user interfaces to improve maintainability and user experience.
- Managing development through Git-based workflows, iterative feedback cycles, and continuous feature delivery.
- Optimizing application performance, accessibility, and cross-device compatibility for real-world usage.

Projects

Smart Asset Tracking System

2025

- Engineered a real-time asset tracking platform leveraging smartphones as IoT nodes, eliminating dependency on dedicated GPS hardware.
- Developed continuous background location tracking and geofencing capabilities optimized for battery efficiency.
- Architected a low-latency synchronization pipeline using Firebase Realtime Database, achieving sub-500ms updates.
- Built a centralized dashboard for real-time route visualization and multi-asset monitoring.
- Applied Isolation Forest and Random Forest models achieving over 85% anomaly detection accuracy.
- Implemented automated alerting workflows for suspicious movement detection.

Data Wiping and Secure Erasure Tool

Aug 2025 – Present

- Developed a security-focused data sanitization platform ensuring irreversible deletion of sensitive information from storage devices.
- Implemented DoD 5220.22-M and Gutmann-compliant overwrite algorithms to prevent data recovery.
- Optimized wipe operations using multithreading and concurrent I/O processing.
- Built a React-based dashboard for real-time monitoring, reporting, and operational control.
- Integrated centralized logging and audit trails to improve compliance and traceability.
- Automated secure deletion workflows, improving operational efficiency by approximately 60%.

Student Database Management System

Feb 2025 – Apr 2025

- Designed a hybrid database architecture using MySQL and MongoDB for academic record management.
- Implemented authentication and role-based access control mechanisms for secure multi-user access.
- Optimized retrieval performance through indexed search, filtering, and sorting techniques.
- Developed modular software components following software engineering best practices.
- Built a Java Swing interface supporting CRUD operations and advanced search functionality.

Achievements

- CGPA: 9.33/10 – Among the top-performing students in the Information Technology program.
- Active problem solver on LeetCode with regular practice in Data Structures and Algorithms.
- Interested in Hackathons, Open Source Contributions, and Competitive Programming.